

6 (a) energy transferred from source / changed from some form to electricalM1
 per unit charge (to drive charge round a complete circuit) A1 [2]

(b) and power in $R = I^2 X$ M1
 $E = I (X + r)$ M1
 power in cell = EI and algebra clear leading to ratio = $X / (X + r)$ A1 [3]

Page 4	Mark Scheme: Teachers' version	Syllabus
	GCE A/AS LEVEL – October/November 2009	9702

- (c) (i) 1.4 W
0.40 Ω (allow $\pm 0.05 \Omega$)
- (ii) current in circuit = $\sqrt{1.4/0.4} = 1.87 \text{ A}$ C1
1.5 = 1.87 ($r + 0.40$) C1
 $r = 0.40 \Omega$ A1 [3]
- (d) either less power lost / energy wasted / lost
or greater efficiency (of energy transfer) B1 [1]

[Total: 11]

- 7 (a) deviation shown correctly B1 [1]